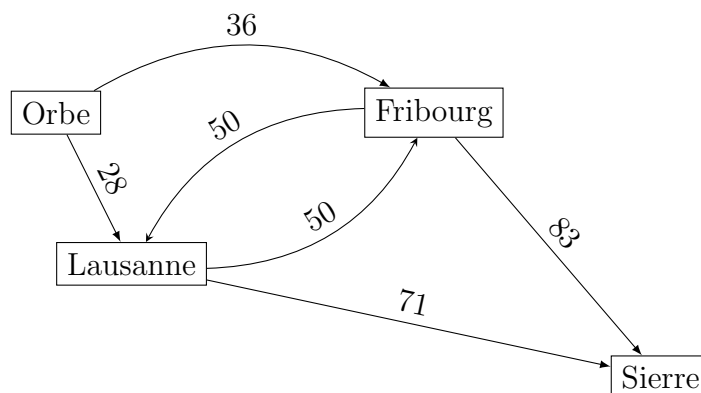


Modeling: shortest path

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Solution of the practice quiz

1. We first model the problem as a transshipment problem.



First of all, we denote Orbe (O), Lausanne (L), Fribourg (F) and Sierre (S). A transshipment problem is written

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}.$$

subject to the flow conservation constraints,

$$\text{div}(x)_i = s_i \quad \forall i \in \mathcal{N},$$

and the bounds on the flows:

$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \quad \forall (i,j) \in \mathcal{A}.$$

In our case,

- $\mathcal{N} = \{O, L, F, S\}$ is the set of nodes,
- $\mathcal{A} = \{(O, L), (O, F), (F, S), (F, L), (L, F), (L, S)\}$ is the set of arcs.
- Costs: $c_{ij} \quad \forall (i, j) \in \mathcal{A}$ is the travel time associated with each arc.
- Supply: as we calculate the shortest path from Orbe to Sierre, we send one unit of flow from Orbe, and consume one unit of flow in Sierre. Therefore,

$$s_O = 1, s_L = 0, s_F = 0, s_S = -1.$$

- Bounds: we impose the flows to be non negative, so that no arc is used in the reverse direction. We do not impose any upper bound here.

The transshipment problem is

$$\min_{x \in \mathbb{R}^6} 28x_{O,L} + 36x_{O,F} + 50x_{L,F} + 83x_{F,S} + 71x_{L,S}$$

subject to

$$\begin{aligned} x_{O,F} + x_{O,L} &= 1 \\ -x_{F,S} - x_{L,S} &= -1 \\ x_{F,L} + x_{F,S} - x_{O,F} - x_{L,F} &= 0 \\ x_{L,S} + x_{L,F} - x_{F,L} - x_{O,L} &= 0 \\ x_{i,j} &\geq 0, \quad \forall (i, j) \in \mathcal{A}. \end{aligned}$$

2. In order to solve it with the simplex algorithm, we first transform it so that the right hand side is non negative:

$$\min_x \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\begin{aligned} x_{O,F} + x_{O,L} &= 1 \\ x_{F,S} + x_{L,S} &= 1 \\ x_{F,L} + x_{F,S} - x_{O,F} - x_{L,F} &= 0 \\ x_{L,S} + x_{L,F} - x_{F,L} - x_{O,L} &= 0 \\ x_{i,j} &\geq 0, \quad \forall (i, j) \in \mathcal{A}. \end{aligned}$$

Phase I: we introduce 4 auxiliary variables x_1, x_2, x_3, x_4 to obtain the auxiliary problem

$$\min x_1 + x_2 + x_3 + x_4$$

subject to

$$\begin{aligned} x_{O,F} + x_{O,L} + x_1 &= 1 \\ x_{F,S} + x_{L,S} + x_2 &= 1 \\ x_{F,L} + x_{F,S} - x_{O,F} - x_{L,F} + x_3 &= 0 \\ x_{L,S} + x_{L,F} - x_{F,L} - x_{O,L} + x_4 &= 0 \\ x_{i,j} &\geq 0, \quad \forall (i,j) \in \mathcal{A} \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

The first tableau is:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	x_1	x_2	x_3	x_4	
1	1	0	0	0	0	1	0	0	0	1
0	0	0	1	0	1	0	1	0	0	1
-1	0	1	1	-1	0	0	0	1	0	0
0	-1	-1	0	1	1	0	0	0	1	0
0	0	0	-2	0	-2	0	0	0	0	-2

The smallest index corresponding to a negative reduced cost is $p = 4$: $x_{F,S}$ enters the basis. The index corresponding to the smallest distance to the constraint $x_{F,S} \geq 0$ is $q = 3$: x_3 leaves the basis. The pivot is $(3, 4)$.

After pivoting the tableau, we obtain:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	x_1	x_2	x_3	x_4	
1	1	0	0	0	0	1	0	0	0	1
1	0	-1	0	1	1	0	1	-1	0	1
-1	0	1	1	-1	0	0	0	1	0	0
0	-1	-1	0	1	1	0	0	0	1	0
-2	0	2	0	-2	-2	0	0	2	0	-2

The smallest index corresponding to a negative reduced cost is $p = 1$: $x_{O,F}$ enters the basis. The indices corresponding to the smallest distance to the constraint $x_{O,F} \geq 0$ are 1 and 2. We choose $q = 1$: x_1 leaves the basis. The pivot is $(1, 1)$.

After pivoting the tableau, we obtain:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	x_1	x_2	x_3	x_4	
1	1	0	0	0	0	1	0	0	0	1
0	-1	-1	0	1	1	-1	1	-1	0	0
0	1	1	1	-1	0	1	0	1	0	1
0	-1	-1	0	1	1	0	0	0	1	0
0	2	2	0	-2	-2	2	0	2	0	0

The smallest index corresponding to a negative reduced cost is $p = 5$: $x_{L,F}$ enters the basis. The indices corresponding to the smallest distance to the constraint $x_{L,F} \geq 0$ are 2 and 4. We choose $q = 2$: x_2 leaves the basis. The pivot is $(2, 5)$.

After pivoting the tableau, we obtain:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	x_1	x_2	x_3	x_4	
1	1	0	0	0	0	1	0	0	0	1
0	-1	-1	0	1	1	-1	1	-1	0	0
0	0	0	1	0	1	0	1	0	0	1
0	0	0	0	0	0	1	-1	1	1	0
0	0	0	0	0	0	0	2	0	0	0

The last tableau is optimal since all reduced costs are non negative and the optimal value is zero. Therefore, we have identified a feasible solution for the initial problem: $x_{O,F} = 1$, $x_{O,L} = 0$, $x_{F,L} = 0$, $x_{F,S} = 1$, $x_{L,F} = 0$, $x_{L,S} = 0$, which corresponds to the path Orbe, Fribourg, Sierre.

We notice that the auxiliary variable x_4 is basic. But we cannot pivot it out of the basis, since all pivots corresponding to non basic variables from the original problem are zero. Indeed, it corresponds to a redundant constraint.

We remove the corresponding row and column to obtain:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	x_1	x_2	x_3	
1	1	0	0	0	0	1	0	0	1
0	-1	-1	0	1	1	-1	1	-1	0
0	0	0	1	0	1	0	1	0	1
0	0	0	0	0	0	0	2	0	0

There are no auxiliary variables left in the basis anymore, and thus we remove the corresponding columns from the tableau.

Furthermore, to begin Phase II of the algorithm, we recalculate the reduced costs and the optimal value to obtain:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	
1	1	0	0	0	0	1
0	-1	-1	0	1	1	0
0	0	0	1	0	1	1
0	42	100	0	0	-62	-119

The smallest index corresponding to a negative reduced cost is $p = 6$: $x_{L,S}$ enters the basis. The index corresponding to the smallest distance to the constraint $x_{L,S} \geq 0$ is $q = 2$: $x_{L,F}$ leaves the basis. The pivot is $(2, 6)$.

After pivoting the tableau we obtain:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	
1	1	0	0	0	0	1
0	-1	-1	0	1	1	0
0	1	1	1	-1	0	1
0	-20	38	0	62	0	-119

The smallest index corresponding to a negative reduced cost is $p = 2$: $x_{O,L}$ enters the basis. The indices corresponding to the smallest distance to the constraint $x_{O,L} \geq 0$ are 1 and 3. We choose $q = 1$: $x_{O,F}$ leaves the basis. The pivot is $(1, 2)$.

After pivoting the tableau we obtain:

$x_{O,F}$	$x_{O,L}$	$x_{F,L}$	$x_{F,S}$	$x_{L,F}$	$x_{L,S}$	
1	1	0	0	0	0	1
1	0	-1	0	1	1	1
-1	0	1	1	-1	0	0
20	0	38	0	62	0	-99

The last tableau is optimal since all reduced costs are non negative.
Thus the optimal solution is:

$$x_{O,F} = 0,$$

$$x_{O,L} = 1,$$

$$x_{F,L} = 0,$$

$$x_{F,S} = 0,$$

$$x_{L,F} = 0,$$

$$x_{L,S} = 1,$$

and corresponds to the path Orbe, Lausanne, Sierre. Note that the solution is integer, although it has not been explicitly imposed. The optimal travel time is 99.